Feeding in Preterm and Feed Intolerance

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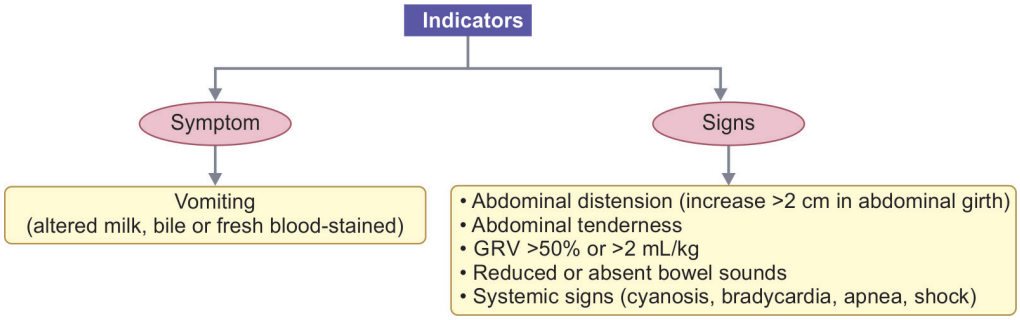
Introduction

Nutritional management in preterm infants plays a vital role not only for adequate postnatal growth, but also for better long-term neurodevelopment outcome. However, there is wide variation of preterm-feeding practices among different neonatal unit, more so because of lack of universal feeding guidelines. Hence, there is need for best practice guidelines, especially for these preterm infants as it is well studied that early initiation of feed is not associated with feed intolerance (FI) and early aggressive enteral nutrition is best and safest way to deliver nutrition in preterm infants.

Feed Intolerance

Definition

- ☑ There is no universally accepted definition for FI. The most comprehensive definition was given by Moore and Wilson who defined FI as the "inability to digest enteral feedings presented as gastric residual volume (GRV) of >50%, abdominal distension or emesis or both, and the disruption of the patient's feeding plan."
- ☑ In most cases, FI represents a benign condition related to the immaturity of gastrointestinal function. Inability to differentiate manifestations of ileus of prematurity from those of early (Stage I) necrotizing enterocolitis (NEC) is the single most important reason for frequently withholding enteral nutrition in preterm neonates and unnecessary withholding feed leads to postnatal growth retardation and possible developmental delay.



Key Points

Quality of gastric residuals	Blood-stained or hemorrhagic residuals, more than bilious residuals, seem to correlate with NEC. Hemorrhagic residue may occur because of injury by feeding tube. Green gastric residues (<2–3 mL) should not slow down the increments of feed volumes in absence of other clinical signs and symptoms.
Refeed or discard gastric residuals	Refeeding supported in adults but no evidence in neonates.
Continuous positive airway pressure (CPAP) belly/noninvasive respiratory support	May cause benign abdominal distension. No evidence to indicate that it is a risk factor for NEC. Usually relieved by simple maneuvers such as prone positioning the neonate or evacuation of air from the stomach before each feed.
Role of glycerin suppositories	No clear benefit as they do not have any effect on the right colon and the small bowel.
Vomiting	Frequency, amount, and color of vomiting that indicates feed intolerance in preterm neonates is not well specified. Bile or blood-stained vomitus should be investigated.
Gastroesophageal reflux and apnea	No clear temporal relationship.

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Investigations	
Laboratory markers	Persistent metabolic acidosis, decreasing platelet count, leukopenia, hyponatremia, and increasing blood glucose level on several successive days might predict a developing NEC
Occult blood in stool	Published evidence suggests that this test is not useful as diagnostic or screening tools. Gross blood in stool, however, needs to be investigated.
Abdominal X-rays	Pneumatosis intestinalis is the most specific radiographic finding for NEC. Other highly suggestive findings include portal venous gas, pneumoperitoneum, and fixed loops of bowel in serial films.
USG abdomen	Advantage of early detection of NEC compare to X-ray abdomen. Signs of NEC in USG: ☑ Increase in gut wall thickening >2.6 mm ☑ Abnormal bowel wall echogenic pattern ☑ Increase in bowel wall vascularity ☑ Minimal free fluid in peritoneal cavity ☑ Along with classical findings of NEC such as pneumatosis intestinalis and portal venous gas
Postnatal superior mesenteric artery (SMA) Doppler indices	No clear role in predicting feed intolerance and NEC in preterm small for gestational age (SGA) babies with absent or reverse end-diastolic flow (AREDF).

Key Points

Color Chart for Vomiting/Gastric Residuals Assessment

Differential Diagnosis of Feeding Intolerance

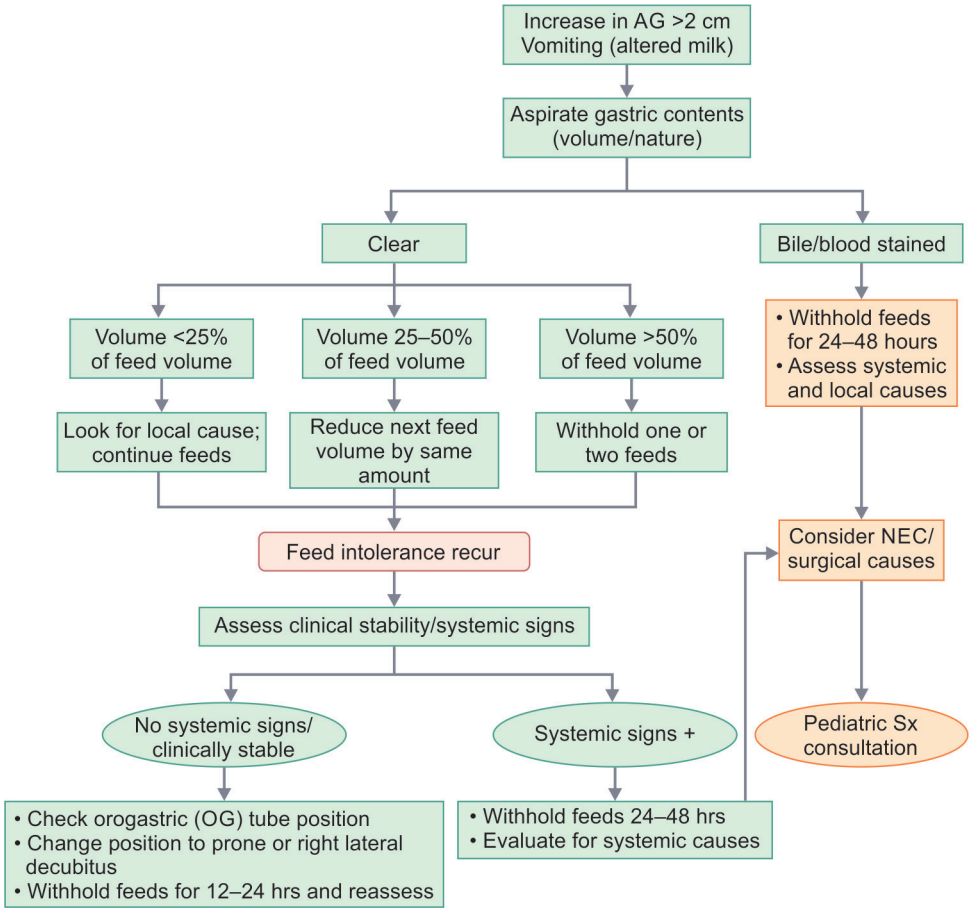
	Clinical feature/additional information
Physiological	Hemodynamically stable neonate, investigation normal
Sepsis	Ileus without abdominal wall erythema/tenderness
NEC	Abdominal distention, emesis, apnea, hemodynamic instability, hyponatremia, thrombocytopenia, and classical abdomen X-ray findings
Spontaneous intestinal perforation (SIP)	Gasless abdomen or asymptomatic pneumoperitoneum; clinical features of NEC may be present; and bluish–gray discolored abdomen is common
Gastroesophageal reflux disease (GERD)	Poor weight gain, apnea, and irritability
Metabolic	Galactosemia
Endocrinal	Thyroid disorder
Surgical conditions	Intestinal obstruction, malrotation, perforation, and intussusception
Allergic colitis	Well-appearing neonate with abdominal distention and bloody stool
Infectious enterocolitis	Presence of loose stools (may be viral or bacterial etiology)

Drugs that delay gastric emptying: Theophylline and caffeine, indomethacin, and mydriatics

Optimal nutrition during the neonatal period is essential for growth and development throughout infancy and into childhood. One of the most challenging tasks is to *adequately and safely* provide nutrition to preterm infants.

Best choice of milk	Mothers own milk → NA → Pasteurized donor human milk → NA → Formula milk
Role of trophic feeding	Should ideally be started within 24 hours of life once hemodynamically stable. Current evidence supports shorter duration trophic feed for shorter time to achieve full feed and early discharge. Colostrum oral swabbing may be considered if feeding is contraindicated or in addition to trophic feeding for its immunological benefits.
Slow versus rapid advancement of feeding	Daily feed volumes are to be increased by 18–40 mL/kg in stable preterm very low birth weight (VLBW) infants.
Routine prefeed abdominal circumference and/or gastric aspiration	Not recommended in the absence of other signs of feed intolerance.
Best route of tube feeding	Intragastric route of tube feeding is preferred over transpyloric route Orogastric tube feeding is preferred over nasogastric tube feeding
Continuous versus bolus feeding	Intermittent bolus feeding is more physiological and is preferred as routine strategy. Continuous feeding is only used in special circumstances.
2 hourly versus 3 hourly feeding	☑ No clinically important differences ☑ Extremely low birth weight (ELBW) (≤1,000 g birth weight) babies should be fed every 2 hourly ☑ >1,000 g birth weight infants may be fed every 2 hourly or 3 hourly
Feeding in SGA/AREDF	For neonates <29 weeks gestation and/or <1,000 g birth weight, increased duration of minimal enteral nutrition and slow advancement are advised as these neonates are at high risk of necrotizing enterocolitis.
Role of human milk fortifier	Multinutrient fortification of breast milk can be initiated in preterm LBW infants with birth weight <1,800 g and receiving enteral feeds of at least 50–80 mL/kg/day. Individualized fortification (targeted or adjusted) is desirable.
Role of supplementing docosahexaenoic acid (DHA)	Routine supplementation is not recommended.
Vitamin A supplementation	Routine oral or intramuscular supplementation of vitamin A is not recommended.
Role of iron supplementation	Oral iron supplements in a daily dose of 2–4 mg/kg/day of elemental iron is recommended from 2 to 4 weeks of life to 12 months of age.
Role of probiotics	Multistrain probiotics containing lactobacilli spp. and *Bifidobacterium* spp. may be initiated in preterm LBW infants from as early as day 1 of life and continued until 36–37 weeks postmenstrual age or discharge (whichever is earlier), in neonatal units with high baseline incidence of necrotizing enterocolitis. However, there is limited data in extremely low birth infants, who are at highest risk of necrotizing enterocolitis.
Role of vitamin D supplementation	VLBW infants should be given vitamin D supplements at a dose ranging from 400 IU to 700 IU per day from the day of reaching full enteral feeds to 6 months to 1 year of age.

Primary aim is careful evaluation in order to differentiate physiological conditions from more dreadful conditions like NEC.



Take Home Message

- ☑ Feed intolerance often over diagnosed.
- ☑ Identify and treat any underlying cause.
- ☑ Avoid excuses to withhold feeding. Instead find reason to continue.

Further Reading

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